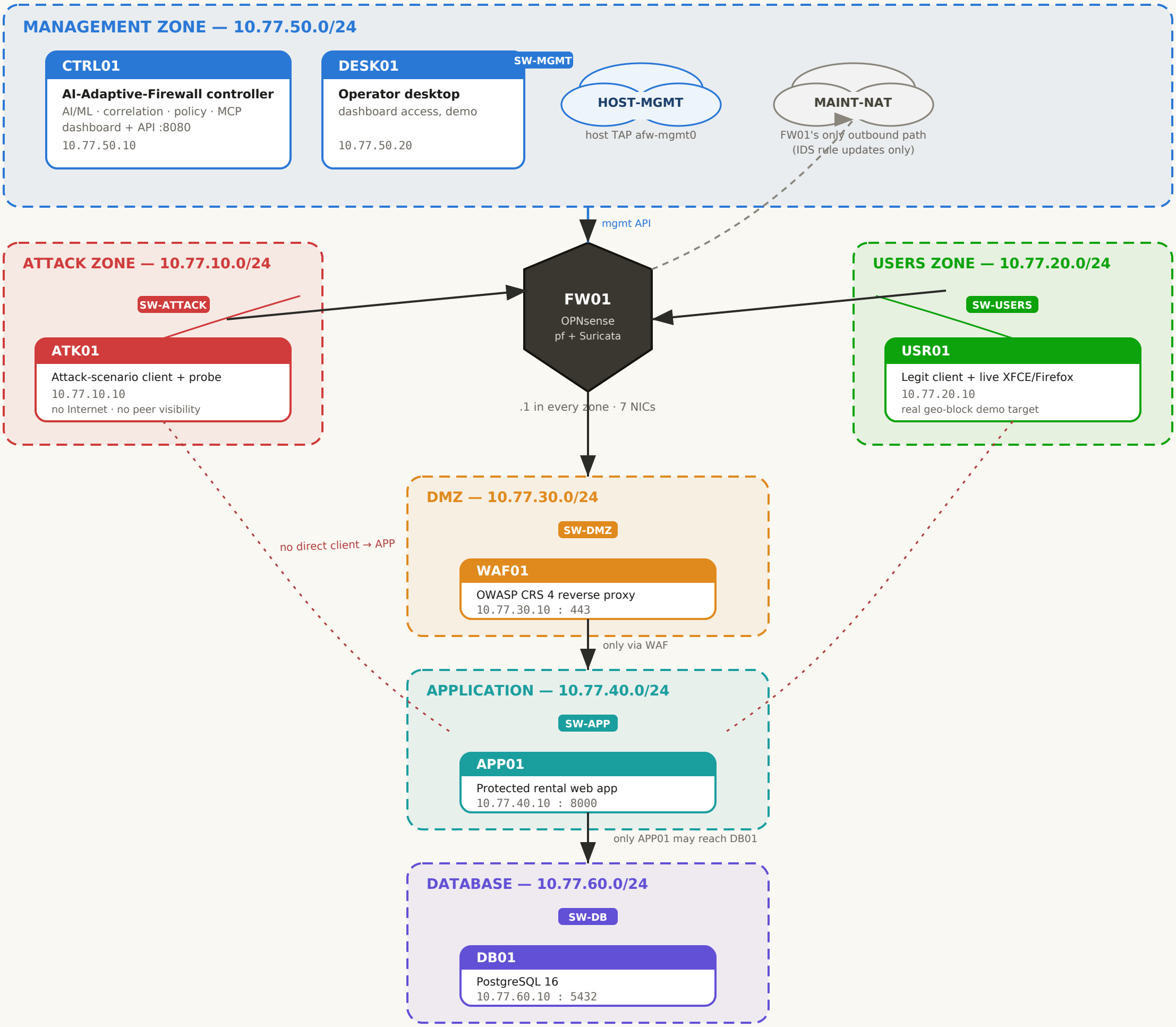
# AI-Adaptive-Firewall — lab topology

Real GNS3 network. Every box is a running VM; every arrow is a traffic path enforced by FW01's firewall policy, not a diagram convention.




## Isolation, verified by code (`afw lab verify`)

- ✓ ATTACK and USERS can reach only WAF01:443 through FW01 — never each other, never APP01 or DB01 directly.
- ✓ APP01 is the only host permitted to reach DB01:5432. The database initiates no outbound traffic of its own.
- ✓ The management plane (CTRL01, DESK01) sits on its own host-only zone, reachable by clients for nothing but telemetry ingestion.
- ✓ MAINT-NAT is FW01's only outbound Internet path, used solely for IDS rule updates — kept off every data zone.

**LEGEND**

Management · Attack (client) · Users (client) · DMZ · Application · Database · Firewall (FW01)

7 functional VMs (CTRL01, FW01, WAF01, APP01, DB01, USR01, ATK01) + 6 zone switches + 2 host connectors = 16 GNS3 nodes, 15 links.

Authoritative source: lab/gns3/topology.yaml · canvas generated by afw lab layout · verified live by afw lab verify.